Lessons from 33 kV cable joint failures, 2026

Summary

Seven 33 kV cable joints failed in service between January and September 2026. The failed joint teardown of each one was part of the cable accessory failure analysis of the field service team.

What the teardowns found

Finding	Joints	Lesson
Cable screen cut back too far during joint fitting	3	Measure the cut-back with the gauge of the jointing procedure
Moisture under the outer sleeve	2	Seal the sleeve ends before rain, not after
No pre-energising tests recorded	2	No energising without an insulation resistance test and a withstand test

Lessons for every field service team

Installation quality records matter: 4 of the 7 joints had no signed job record, so the jointer and the date could not be traced. From November 2026 a joint is energised only after its digital job card is signed.

The field service routine adds a visit at 30 days after energising for every new 33 kV joint.

Synthetic sample written for the RPG NeuralMesh demonstration. It is not a record of the RPG Group.